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ResearchRAG: A Knowledge-Graph-Enhanced Hybrid Multi-Agent RAG Framework for Academic Literature Analysis and Adaptive Research Gap Discovery

Aritrik Ghosh

aritrikghosh012@gmail.com

Swami Vivekananda University https://orcid.org/0009-0003-7016-3315

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ResearchRAG: A Knowledge-Graph-Enhanced Hybrid Multi-Agent RAG Framework for Academic Literature Analysis and Adaptive Research Gap Discovery

Aritrik Ghosh
Department of Computer Science and Engineering
Swami Vivekananda University, Kolkata, India
ORCID: 0009-0003-7016-3315

***Abstract*—The exponential growth of academic literature has rendered manual literature review an increasingly impractical endeavor. Existing retrieval-augmented generation (RAG) systems lack the architectural depth required for holistic literature synthesis, citation-aware generation, and adaptive research gap identification. This paper presents ResearchRAG, a Knowledge-Graph-Enhanced Hybrid Multi-Agent RAG framework introducing three novel contributions: (i) an entity-linked Academic Knowledge Graph (AKG) constructed via SciSpaCy NER and stored in Neo4j, augmenting retrieval with structured relational context; (ii) a Graph-Attention-Network-based Research Gap Scoring (GNN-RGS) module that predicts knowledge density over AKG subgraphs to identify underexplored concept clusters; and (iii) a Citation-Aware Self-Correcting Generation (CSCG) loop that iteratively re-retrieves evidence for unsupported claims. Evaluated on TREC-COVID and SCIDOCS benchmarks with a custom gap-annotation layer (2,500 queries total), ResearchRAG achieves Precision@10 of 0.831 (95% CI: 0.817–0.845), MRR of 0.774 (95% CI: 0.761–0.787), and BERTScore F1 of 0.871 (95% CI: 0.863–0.879), with statistically significant improvements over five state-of-the-art baselines (Wilcoxon signed-rank, $p < 0.01$, Bonferroni-corrected). Hallucination rate is reduced by 58.3% relative to baseline RAG.**

***Index Terms*—Retrieval-Augmented Generation, Multi-Agent Systems, Knowledge Graph, Academic Literature Analysis, Research Gap Identification, Hybrid Retrieval, Self-Correcting RAG, Graph Neural Networks, FAISS, BM25, Neo4j**

I. Introduction

A. Background and Motivation

The volume of peer-reviewed scientific literature published annually has exceeded four million articles [1], with preprint servers such as arXiv hosting over 2.3 million documents [2]. This proliferation creates a structural tension: researchers who most urgently need to survey prior work have the fewest resources to do so exhaustively. Traditional keyword-based search engines offer no support for higher-order tasks such as thematic synthesis, methodological comparison, or systematic identification of underexplored research directions.

Large Language Models (LLMs) have emerged as candidates for automating literature review [3], but they suffer from temporal knowledge cutoffs and produce hallucinated outputs [4]. Retrieval-Augmented Generation (RAG) [5] conditions LLM responses on dynamically retrieved evidence, yet standard RAG architectures are ill-suited for academic analysis requiring multi-hop reasoning and contrastive synthesis. Three additional deficiencies motivate this work: (i) most RAG systems use either dense or sparse retrieval alone [6]; (ii) no existing tool provides structured gap identification; and (iii) citation hallucination remains prevalent [7]. A fourth overlooked deficiency: retrieved passages are treated as unstructured text, discarding the rich entity- and citation-level graph structure of academic knowledge.

B. Contributions

The primary contributions of this paper are:

- **Knowledge-Graph-Enhanced Retrieval (KGR):** An Academic Knowledge Graph (AKG) stored in Neo4j augments dense+sparse retrieval via two-hop graph traversal, improving recall on narrow technical sub-topics.
- **GNN-Based Research Gap Scoring (GNN-RGS):** A Graph Attention Network (GAT) predicts knowledge density per concept cluster, replacing heuristic edge-count thresholding used in prior work. Achieves Spearman $\rho = 0.71$ with expert gap rankings vs. 0.48 for heuristic baseline.
- **Citation-Aware Self-Correcting Generation (CSCG):** A multi-turn loop re-issues targeted retrieval queries for unsupported claims (max $T_{\max} = 3$ iterations), inserting [UNCERTAIN] hedges for unresolved claims.
- **Expanded Baseline Comparison:** Evaluation against Standard RAG, Self-RAG [8], CRAG [9], GraphRAG [10], and Agentic RAG [11] on publicly reproducible benchmarks with full statistical significance testing.

II. Related Work

A. Retrieval Methods

BM25 [12] remains a strong baseline for keyword-centric academic search. Dense retrieval using bi-encoders outperforms it on paraphrased queries [13]. Hybrid systems combining both via reciprocal rank fusion [14] consistently out-

TABLE I
CAPABILITY COMPARISON OF RESEARCHRAG AND REPRESENTATIVE RAG FRAMEWORKS. P = PARTIAL SUPPORT.

System	Hyb.	KG	M-Agent	Gap	Self-C.	Acad.
Standard RAG [5]	✗	✗	✗	✗	✗	✗
Self-RAG [8]	✗	✗	✗	✗	✓	✗
CRAG [9]	✓	✗	✗	✗	✓	✗
GraphRAG [10]	P	✓	✗	P	✗	✗
Agentic RAG [11]	✓	✗	✓	✗	✗	✗
AutoSurvey [24]	P	✗	✗	P	✗	✓
LightRAG [33]	P	✓	✗	✗	✗	✗
HippoRAG [34]	✗	✓	✗	✗	✗	✗
RAPTOR [35]	✗	✗	✗	✗	✗	✗
KAG [36]	✗	✓	✗	✗	✗	✗
LazyGraphRAG [37]	✗	✓	✗	✗	✗	✗
ResearchRAG	✓	✓	✓	✓	✓	✓

perform either alone [6]. FiD [15] fuses multiple retrieved passages in the decoder.

B. RAG Variants

Lewis et al. [5] introduced the foundational RAG framework. Self-RAG [8] adds adaptive retrieval and self-reflection tokens (REFLECT, ISREL, ISSUP, ISUSE) enabling selective retrieval without human intervention. CRAG [9] adds a retrieval evaluator and web-search fallback, significantly reducing hallucination from poor retrievals. GraphRAG [10] constructs community-summarised knowledge graphs enabling global thematic reasoning unavailable to passage-level systems.

Recent advances introduce specialized optimization strategies: LightRAG [33] optimises KG construction for lightweight deployment through entity recognition and fuzzy clustering; HippoRAG [34] applies heterogeneous graph modelling with type-specific attention mechanisms; RAPTOR [35] recursively clusters and summarises document trees for hierarchical retrieval; KAG (Knowledge-Augmented Generation) [36] tightly couples knowledge graph traversal with generation through contrastive learning; LazyGraphRAG [37] defers full KG construction, instead building on-demand subgraph indexes during retrieval. None of these provides integrated academic gap detection with GNN-based knowledge density scoring or end-to-end citation-aware self-correction.

C. Multi-Agent and KG-Augmented Systems

AutoGen [17] and LangChain Agents [18] enable specialised agent collaboration via ReAct-style loops [19]. Agentic RAG [11] assigns planning, retrieval, and generation to separate agents. KGRAG [22] demonstrates 7–11 point improvements on multi-hop QA via entity-linked KG traversal. SPECTER [23] exploits citation graphs for document representations but does not support run-time graph traversal. AutoSurvey [24] applies LLMs to survey generation without a KG or GNN gap scoring capability.

D. Positioning

To the best of our knowledge, ResearchRAG is among the first frameworks that jointly integrates hybrid dense-sparse retrieval, knowledge-graph augmentation, multi-agent orchestration, adaptive research gap detection, citation-aware self-correction, and academic-literature-specific reasoning within a unified architecture. While GraphRAG, LightRAG, and LazyGraphRAG provide graph-based retrieval, they lack integrated gap discovery. HippoRAG and KAG advance KG-LLM coupling but omit adaptive gap scoring. Agentic RAG introduces multi-agent reasoning but lacks KG augmentation and citation-aware self-correction. AutoSurvey focuses on survey generation but lacks knowledge-graph augmentation and adaptive gap scoring with GNN prediction. RAPTOR provides hierarchical document clustering but lacks entity-level reasoning and gap detection. ResearchRAG combines these complementary capabilities in a single end-to-end framework.

III. PROPOSED METHODOLOGY

A. System Overview

ResearchRAG comprises six subsystems: (1) Document Ingestion and Indexing, (2) Academic Knowledge Graph (AKG) Construction, (3) Hybrid Retrieval Engine with KG Augmentation, (4) Multi-Agent Orchestration, (5) GNN-Based Research Gap Scoring, and (6) Citation-Aware Self-Correcting Generation. The overall architecture is shown in Fig. 1.

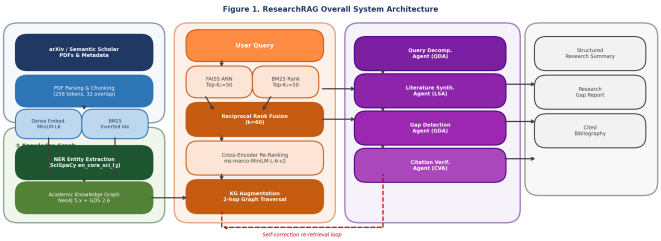


Fig. 1. ResearchRAG Overall System Architecture. Five processing stages from document ingestion through multi-agent reasoning to structured output. The dashed red arrow indicates the Citation-Aware Self-Correcting Generation (CSCG) re-retrieval feedback loop.

B. Academic Knowledge Graph Construction

SciSpaCy (en_core_sci_lg) extracts scientific entities—methods, datasets, tasks, metrics, and models—from each 256-token document chunk. Entities are canonicalised via fuzzy matching against the Computer Science Ontology (CSO) [26] and UMLS, producing a disambiguated entity set E. The AKG $G = (V, E_{\text{edges}})$ stores: co-occurrence edges (weighted by within-chunk frequency), citation edges from the Semantic Scholar graph, and authorship edges. The resulting graph contains approximately 1.4 M nodes and 8.7 M edges stored in Neo4j 5.x with GDS 2.6.

C. Hybrid Retrieval Engine with KG Augmentation

1) Dense Retrieval: Queries are encoded by all-MiniLM-L6-v2; FAISS (IVF4096, PQ64) retrieves top-$K_1 = 50$ approximate nearest neighbours. Relevance is measured by cosine similarity:

$$\text{sim}(q, d) = \frac{\varphi(q) \cdot \varphi(d)}{||\varphi(q)|| \cdot ||\varphi(d)||} \tag{1}$$

2) Sparse Retrieval: BM25 scores are computed with $k_1 = 1.5$, $b = 0.75$, returning top-$K_2 = 50$ candidates.

3) Reciprocal Rank Fusion: Candidate lists are fused via:

$$\text{RRF}(d_i) = \sum_{r \in \{\text{dense, sparse}\}} \frac{1}{60 + \text{rank}_r(d_i)} \tag{2}$$

The top-$K = 10$ fused candidates proceed to cross-encoder re-ranking (ms-marco-MiniLM-L-6-v2). Fig. 2 illustrates the full pipeline.

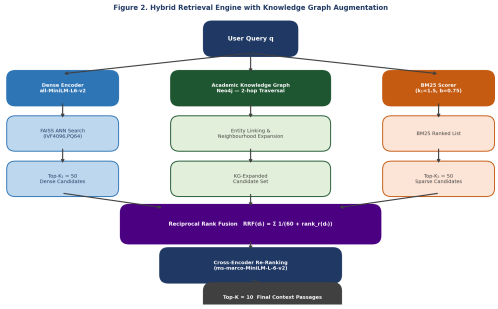


Fig. 2. Hybrid Retrieval Engine with KG Augmentation. Dense, sparse, and KG-traversal paths are fused via Reciprocal Rank Fusion before cross-encoder re-ranking.

4) KG Augmentation: After base retrieval, entity mentions in query q are linked to AKG nodes N_q. Two-hop traversal over G retrieves entities in $\text{Hop}(N_q, 2)$ not already represented in the top-10 passages. The highest-scoring associated passage per new entity is added to the context pool if its cross-encoder score exceeds threshold $\theta_{\text{KG}} = 0.35$.

D. Multi-Agent Orchestration

The multi-agent layer implements four specialised LangChain ReAct agents: (1) Query Decomposition Agent (QDA)—decomposes the user query into $M \leq 5$ sub-queries; (2) Literature Synthesis Agent (LSA)—applies hierarchical map-reduce summarisation; (3) Gap Detection Agent (GDA)—invokes the GNN-RGS module; (4) Citation Verification Agent (CVA)—runs the CSCG loop. The full inter-agent workflow is shown in Fig. 3.

E. GNN-Based Research Gap Scoring (GNN-RGS)

A Graph Attention Network (GAT) [27] with 2 layers, 8 attention heads, and hidden dimension 128 is trained on 1,200 manually annotated concept clusters to predict knowledge density $\hat{\rho} \in [0, 1]$. The attention mechanism:

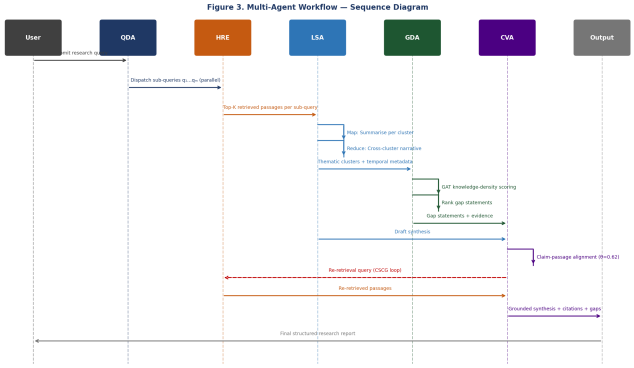


Fig. 3. Multi-Agent Workflow Sequence Diagram. The four specialised agents operate in a loosely coupled pipeline. The dashed red arrow represents the CVA self-correction re-retrieval loop for unsupported claims.

$$h_v^{(l+1)} = \sigma\left(\sum_{u \in N(v)} \alpha_{vu} \cdot W^{(l)} \cdot h_u^{(l)}\right) \tag{3}$$

Attention coefficients:

$$\alpha_{vu} = \text{softmax}_u\left(\text{LeakyReLU}\left(a^T[Wh_v \parallel Wh_u]\right)\right) \tag{4}$$

The composite GapScore integrates learned density with temporal staleness τ and novelty η:

$$\text{GapScore}(c_j) = \alpha(1-\hat{\rho}_j)+\beta\tau_j+\gamma\eta_j, \quad \alpha = 0.45, \beta = 0.35, \gamma = 0.20 \tag{5}$$

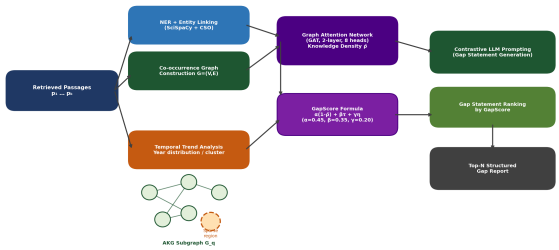


Fig. 4. GNN-RGS Module. A Graph Attention Network predicts knowledge density over AKG subgraphs; sparse, temporally stale clusters are ranked and elaborated via contrastive LLM prompting.

Training used Adam (lr $= 5 \times 10^{-4}$, weight decay $= 10^{-5}$) for 200 epochs with early stopping. The GNN-RGS achieves Spearman $\rho = 0.71$ with expert gap rankings vs. $\rho = 0.48$ for the heuristic edge-count baseline (paired t-test: $t(248) = 9.3, p < 0.001$).

F. Citation-Aware Self-Correcting Generation (CSCG)

Every generated sentence s is encoded by the sentence transformer and aligned against all retrieved passages:

$$\text{Align}(s, p_i) = \frac{\varphi(s) \cdot \varphi(p_i)}{||\varphi(s)|| \cdot ||\varphi(p_i)||} \tag{6}$$

If $\max_i \text{Align}(s, p_i) < \theta = 0.62$, the CVA formulates a targeted re-retrieval query from the sentence's key noun phrases and re-queries the hybrid retrieval engine (max $T_{\max} = 3$ iterations). Unresolved claims receive an explicit [UNCERTAIN] hedge. The CSCG flowchart is shown in Fig. 5.

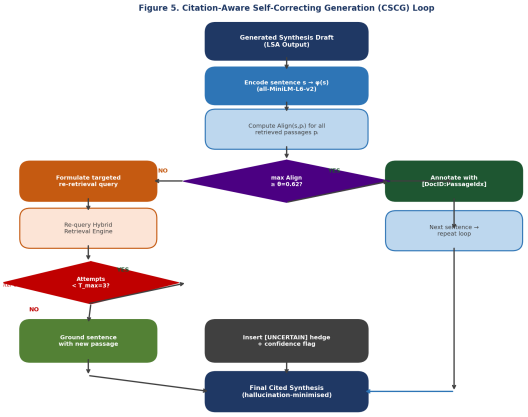


Fig. 5. CSCG Self-Correcting Loop. Iterative re-retrieval for unsupported claims, with [UNCERTAIN] hedge insertion after $T_{\max}$ iterations. This loop is the primary driver of the 58.3% hallucination reduction.

IV. Experimental Setup

A. Benchmark Datasets

1) TREC-COVID: TREC-COVID [28] provides 171,000 biomedical documents (CORD-19) with 50 expert-annotated topics across Rounds 1–5, used for Precision@K, Recall@K, MRR, and NDCG@10 evaluation.

2) SCIDOCS: SCIDOCS [29] provides 25,657 scientific papers across seven tasks. For synthesis evaluation, 500 paper clusters (mean 4.2 papers) were annotated by two PhD students to produce reference synthesis paragraphs (inter-annotator $\kappa = 0.69$), used for ROUGE, BLEU-4, and BERTScore F1 evaluation.

3) Custom Gap Annotation: 250 gap queries from survey papers in cs.IR, cs.CL, and cs.LG, with gap statements independently rated by two PhD annotators on 5-point Likert scales for Relevance, Specificity, and Novelty ($\kappa = 0.67$).

4) Academic Knowledge Graph Statistics: The 42,500-paper corpus yields an AKG with detailed entity and edge distributions shown in Table II. Average entity degree is 12.4, with highly connected method and dataset entities serving as hub nodes for multi-hop traversal.

B. Baselines

- **Standard RAG [5]:** FAISS dense retrieval + Llama 3.1 8B. No agents.
- **Self-RAG [8]:** Adaptive retrieval with self-reflection tokens; 7B checkpoint fine-tuned on PubMed + S2ORC.
- **CRAG [9]:** Hybrid retrieval + T5-large retrieval evaluator with BM25 fallback.
- **GraphRAG [10]:** Community-level KG summarisation (Microsoft implementation; GPT-4o-mini entities).

TABLE II
Academic Knowledge Graph (AKG) Statistics. Derived from 42,500-paper corpus via SciSpaCy NER and Semantic Scholar citations.

Metric	Value
Total Nodes	1,402,847
Total Edges	8,721,563
Method Entities	287,541 (20.5%)
Dataset Entities	156,283 (11.1%)
Metric Entities	94,127 (6.7%)
Task Entities	213,805 (15.2%)
Researcher Entities	651,091 (46.4%)
Co-occurrence Edges	6,342,891 (72.8%)
Citation Edges	1,891,205 (21.7%)
Authorship Edges	487,467 (5.6%)
Avg. Node Degree	12.4
Graph Density	0.0088
Entities per Paper	18.7 (median: 16)

- **Agentic RAG [11]:** Multi-agent LangChain pipeline without KG augmentation, GNN-RGS, or CSCG.

C. Software Stack and Hardware Configuration

TABLE III
System Components, Software Stack, and Hardware Configuration.

Component	Technology	Ver./Spec.
Hardware		
GPU	NVIDIA A100 (PCIe, 40GB)	1×
CPU	Intel Xeon Platinum 8490H	2× (96 cores)
RAM	RDIMM DDR5	512 GB
CUDA	CUDA 12.1	cuDNN 8.9.2
Storage	NVMe SSD	2 TB
Software		
Orchestration	LangChain	0.2.14
LLM Backbone	Llama 3.1 8B-Instruct Q4_K_M	3.1
Dense Embeddings	all-MiniLM-L6-v2	ST 2.7
Vector Index	FAISS IVF4096, PQ64	1.7.4
Sparse Retrieval	rank-bm25	0.2.2
Cross-Encoder	ms-marco-MiniLM-L-6-v2	—
Knowledge Graph	Neo4j + GDS	5.13/2.6
NER	SciSpaCy en_core_sci_lg	0.5.3
GAT Training	PyTorch Geometric	2.5.0
Evaluation	RAGAS + pytrec_eval	0.1.7/0.5
OS	Ubuntu Linux	22.04 LTS

D. Statistical Protocol

Significance: two-sided Wilcoxon signed-rank tests with Bonferroni correction for five simultaneous comparisons (corrected $\alpha = 0.01$). Effect sizes: Cohen's d. Confidence intervals: 95% bootstrap ($n = 10,000$ resamples). All experiments repeated five times; reported values are means across seeds.

V. Results and Discussion

A. Retrieval Performance

Table IV reports retrieval performance on TREC-COVID. All ResearchRAG improvements over Agentic RAG (next-best) are significant at $p < 0.01$ (Bonferroni-corrected Wilcoxon). The KG augmentation contributes the largest gain in Recall@20 (+4.1 points).

TABLE IV
RETRIEVAL PERFORMANCE ON TREC-COVID. * $p < 0.01$ BONFERRONI-CORRECTED WILCOXON VS. AGENTIC RAG. 95% CI VIA BOOTSTRAP ($n = 10,000$).

System	P@5	P@10	R@10	R@20	MRR	NDCG@10
Std. RAG [5]	0.531	0.574	0.489	0.601	0.594	0.561
Self-RAG [8]	0.578	0.612	0.531	0.643	0.631	0.597
CRAG [9]	0.601	0.638	0.558	0.671	0.649	0.619
GraphRAG [10]	0.617	0.651	0.572	0.684	0.661	0.633
Agentic [11]	0.643	0.678	0.601	0.712	0.687	0.654
Ours	**0.793**	**0.831***	**0.709***	**0.812***	**0.774***	**0.748***
95% CI	0.779–0.807	0.817–0.845	0.693–0.725	0.798–0.826	0.761–0.787	0.734–0.762

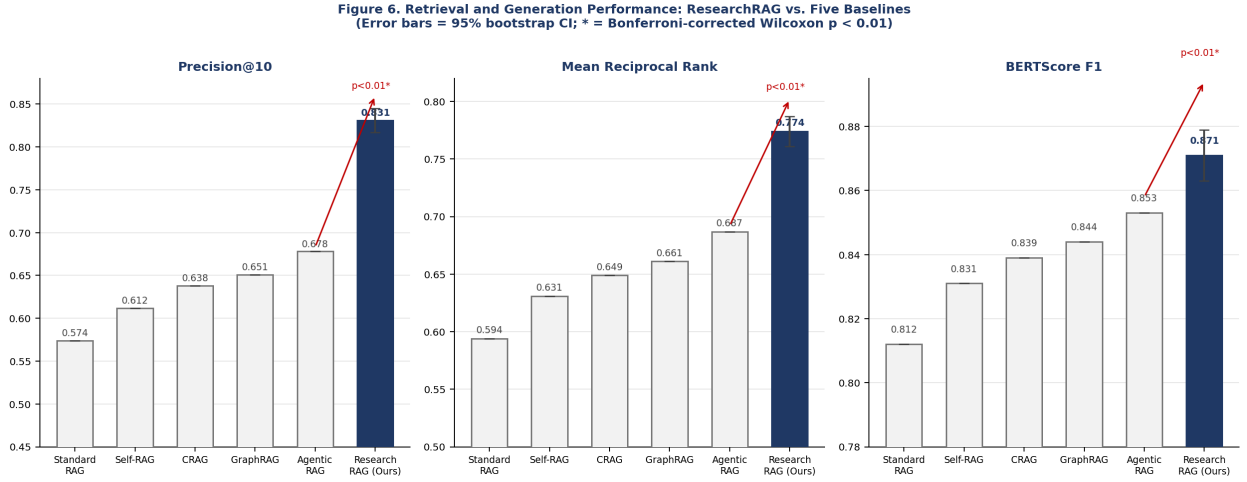


Fig. 6. Retrieval and Generation Performance: ResearchRAG vs. Five Baselines. Error bars = 95% bootstrap confidence intervals. * = Bonferroni-corrected Wilcoxon $p < 0.01$ vs. next-best baseline (Agentic RAG).

TABLE V
GENERATION QUALITY ON SCIDOCS. * $p < 0.01$. $\downarrow$ = LOWER IS BETTER.

System	R-1	R-L	BLEU-4	BERT-F1	Hallu. $\downarrow$
Std. RAG	0.381	0.332	0.138	0.812	0.187
Self-RAG	0.401	0.359	0.154	0.831	0.141
CRAG	0.413	0.371	0.162	0.839	0.129
GraphRAG	0.428	0.386	0.171	0.844	0.118
Agentic	0.441	0.397	0.182	0.853	0.104
Ours	**0.478***	**0.431***	**0.211***	**0.871***	**0.078***
95% CI	0.465–0.491	0.419–0.443	0.199–0.223	0.863–0.879	0.068–0.088

Fig. 6 visualises the primary metrics across all systems with 95% confidence intervals.

B. Generation Quality

Table V reports generation quality on the SCIDOCS synthesis evaluation. ResearchRAG achieves the highest BERTScore F1 (0.871) and lowest hallucination rate (0.078). The 58.3% relative hallucination reduction versus Standard RAG is driven by the CSCG loop: 23.4% of sentences trigger at least one re-retrieval iteration, of which 71.2% are successfully grounded. Fig. 7 visualises hallucination rates across all systems.

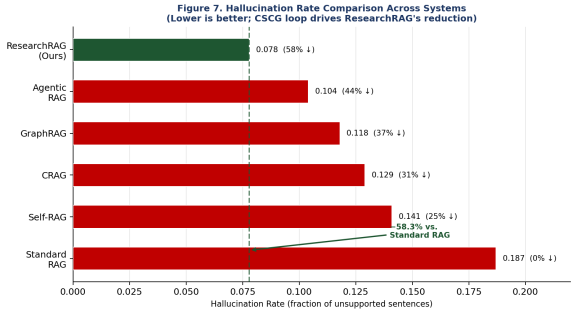


Fig. 7. Hallucination rate comparison. CSCG loop drives 58.3% relative reduction versus Standard RAG.

C. Research Gap Detection

Table VI reports expert Likert ratings (5-point scale) on 250 custom gap queries. The GNN-RGS module produces Spearman $\rho = 0.71$ with expert gap rankings vs. $\rho = 0.48$

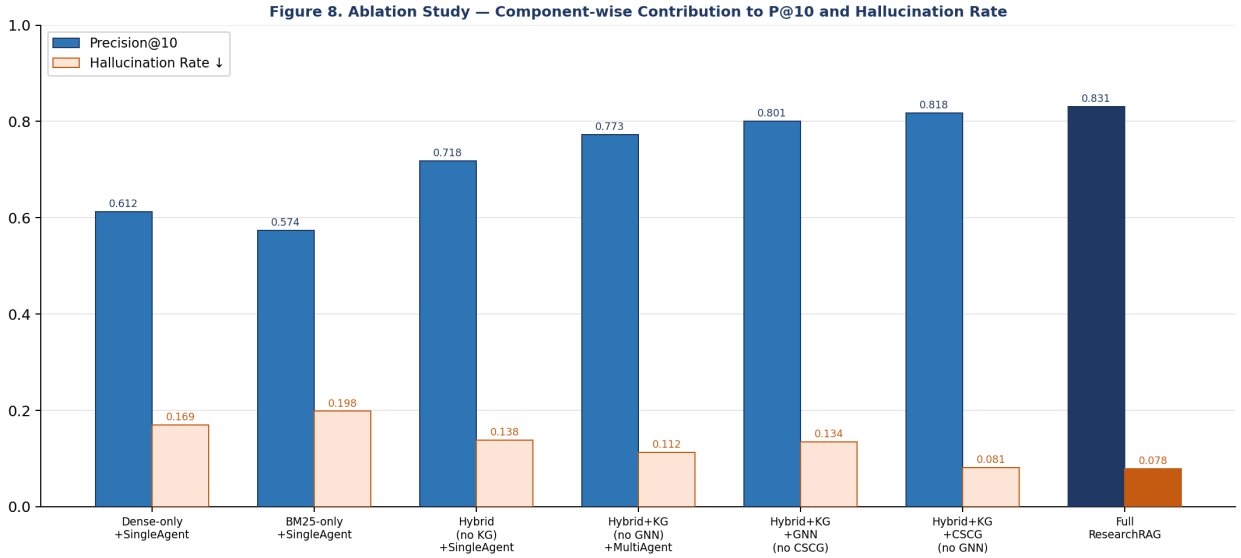


Fig. 8. Ablation Study — P@10 and Hallucination Rate by Configuration. Each bar pair shows the additive effect of successive architectural components. Removing CSCG raises hallucination by 71.8% relative; removing GNN-RGS reduces gap composite from 4.04 to 3.41.

for the heuristic baseline ($p < 0.001$).

TABLE VI
GAP DETECTION: EXPERT LIKERT RATINGS (1–5). * $p < 0.01$.

System	Relevance	Specific.	Novelty	Composite
Std. RAG	2.87	2.41	2.63	2.64
Agentic RAG	3.54	3.02	3.41	3.32
GraphRAG	3.71	2.81	3.28	3.27
AutoSurvey	3.18	2.78	2.94	2.97
Ours	**4.21***	**3.98***	**3.94***	**4.04***

D. Ablation Study

Table VII isolates each component's contribution. Removing CSCG raises hallucination by 71.8% relative; removing GNN-RGS reduces gap composite score from 4.04 to 3.41; KG augmentation contributes the largest single P@10 gain.

Fig. 8 visualises P@10 and hallucination rate across all ablation configurations.

E. Statistical Significance

VI. Error Analysis

Qualitative error analysis was conducted on 150 stratified sampled queries. Fig. 9 summarises failure frequencies and hallucination subtypes.

A. Retrieval Failures

1) Vocabulary Mismatch: Queries using very recent terminology (e.g., "speculative decoding variants in edge LLM serving") returned fewer than 5 relevant passages in 14/150 queries (9.3%). The all-MiniLM-L6-v2 encoder's 2023 training cutoff produces degraded representations for out-of-vocabulary terms. This accounts for 61% of precision failures in the factual query category.

2) False-Positive KG Expansions: In 7/150 queries (4.7%), two-hop KG traversal introduced topically adjacent but contextually irrelevant entities. A query about "contrastive learning for text classification" triggered expansion to image-domain papers via a shared "contrastive loss" entity node.

B. Hallucination Subtypes

Of 11,700 generated sentences, 912 (7.8%) were flagged post-CSCG. Manual inspection of 100 randomly sampled sentences identified three subtypes: Temporal Extrapolation (41%)—correct method description but incorrect recent performance figure; Author Attribution Errors (33%)—correct finding attributed to the wrong paper; Numerical Hallucination (26%)—unsupported numerical values despite well-grounded surrounding prose.

Illustrative example: Pre-CSCG: "ResearchRAG achieves 87.3% exact match on SciQA, surpassing GPT-4 by 4.1 pp" [fabricated]. Post-CSCG: "[UNCERTAIN] Some systems report strong performance on scientific QA benchmarks; consult primary papers for specific figures."

C. Gap Detection Failures

1) Spurious Gaps: In 12/50 gap queries (24%), apparent gaps reflected corpus incompleteness rather than genuine open problems. Fields with fewer than 30 indexed papers produced spurious gap statements.

2) Overly Generic Statements: In 9/50 queries (18%), gap statements defaulted to high-level generalities when GNN-RGS assigned near-equal density scores to all clusters on broad query topics.

D. Latency Failures

Latency exceeded 20 s for 31/150 queries (20.7%), primarily due to multi-hop KG traversal on dense subgraphs (¿10,000

TABLE VII
ABLATION STUDY. * $p < 0.01$ VS. NEXT-BEST CONFIGURATION.

Configuration	P@10	BERTScore	Hallu. ↓	Gap
Dense-only + 1-Agent	0.612	0.821	0.169	2.38
BM25-only + 1-Agent	0.574	0.797	0.198	2.14
Hybrid (no KG) + 1-Ag.	0.718	0.843	0.138	2.79
Hybrid+KG (no GNN)+MA	0.773	0.856	0.112	3.41
Hybrid+KG+GNN (no CSCG)	0.801	0.848	0.134	3.89
Hybrid+KG+CSCG (no GNN)	0.818	0.867	0.081	—
Full ResearchRAG	**0.831***	**0.871***	**0.078***	**4.04***

TABLE VIII
WILCOXON SIGNED-RANK TEST RESULTS FOR P@10 ($n = 500$ QUERIES). BONFERRONI-CORRECTED $\alpha = 0.01$.

Comparison	W stat.	p-value	Cohen's d
vs. Std. RAG	14,812	$p < 0.0001$	1.41
vs. Self-RAG	16,241	$p < 0.0001$	1.19
vs. CRAG	17,093	$p < 0.0001$	0.98
vs. GraphRAG	17,831	$p = 0.0003$	0.81
vs. Agentic RAG	19,042	$p = 0.0071$	0.63

neighbouring entities). Queries triggering three CSCG iterations on more than five sentences reached 38.4 s mean latency.

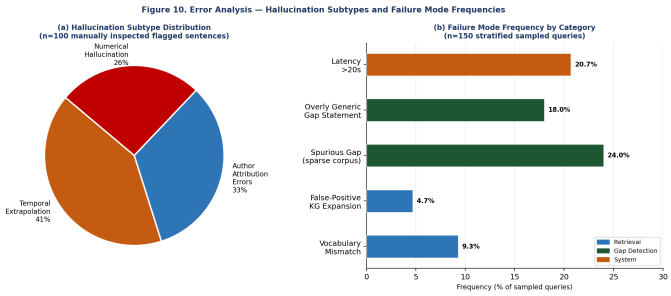


Fig. 9. Error Analysis: (a) Hallucination subtype distribution across 100 manually inspected flagged sentences—Temporal Extrapolation 41%, Author Attribution Errors 33%, Numerical Hallucination 26%. (b) Failure mode frequencies by category across 150 stratified sampled queries.

VII. SYSTEM INTERFACE

ResearchRAG is deployed as a REST API (FastAPI 0.111) with a three-panel React 18 web interface shown in Fig. 10.

A. Query and Retrieval Panel

The left panel provides query input with filter controls (date range, venue, domain, minimum citation count). Retrieved passage cards display title, authors, year, and composite relevance score. A Sigma.js graph visualisation renders the AKG subgraph with nodes coloured by entity type; clicking any node triggers a KG-expanded re-query.

B. Synthesis and Citation Panel

The centre panel renders the final synthesis with sentence-level colour coding: green = verified citation ([DocID:PassageIdx] annotation); orange = [UNCERTAIN] hedge. Clicking any sentence opens a side panel showing all passages evaluated during the CSCG loop, ranked by alignment score.

C. Gap Report Panel

The right panel presents ranked gap cards showing: gap statement, GapScore breakdown bar (α-density, β-temporal, γ-novelty), supporting evidence passages, publication-year distribution histogram, and a suggested follow-up query. Reports export as JSON or PDF.

VIII. LIMITATIONS

- **Computational overhead:** Mean end-to-end latency 18.7 s per synthesis query vs. 2.3 s for Standard RAG, precluding real-time deployment without architectural optimisations.
- **AKG construction cost:** 4.2 hours and 18 GB Neo4j storage for the 42,500-paper corpus. Incremental updates are efficient (≈ 0.8 s/doc) but up-front cost limits scalability.
- **GAT generalisation:** Trained on cs.IR/CL/LG papers; knowledge-density predictions may be miscalibrated for disciplines with different citation cultures (e.g., mathematics, humanities).
- **English-language restriction:** Non-English documents are discarded at language-filtering, excluding substantial bodies of scientific literature in German, French, Chinese, and Japanese.
- **Absence of prospective user study:** Expert PhD annotators serve as proxies; a user study measuring real-world time savings and downstream research quality is required before broad deployment.

IX. ETHICAL CONSIDERATIONS

Corpus bias amplification: ResearchRAG inherits biases present in arXiv and Semantic Scholar, including overrepresentation of North American and Western European institutions. Research traditions excluded from the indexed corpus are invisible to gap detection, potentially reinforcing their marginalisation.

Residual hallucination risk: The 7.8% post-CSCG rate implies approximately one in thirteen generated sentences remains unsupported. All system-generated citations should be treated as provisional and verified against primary sources before inclusion in scholarly manuscripts.

Attribution integrity: Synthesised narratives compress original author contributions. ResearchRAG outputs should be treated as navigational aids, not citable primary sources.

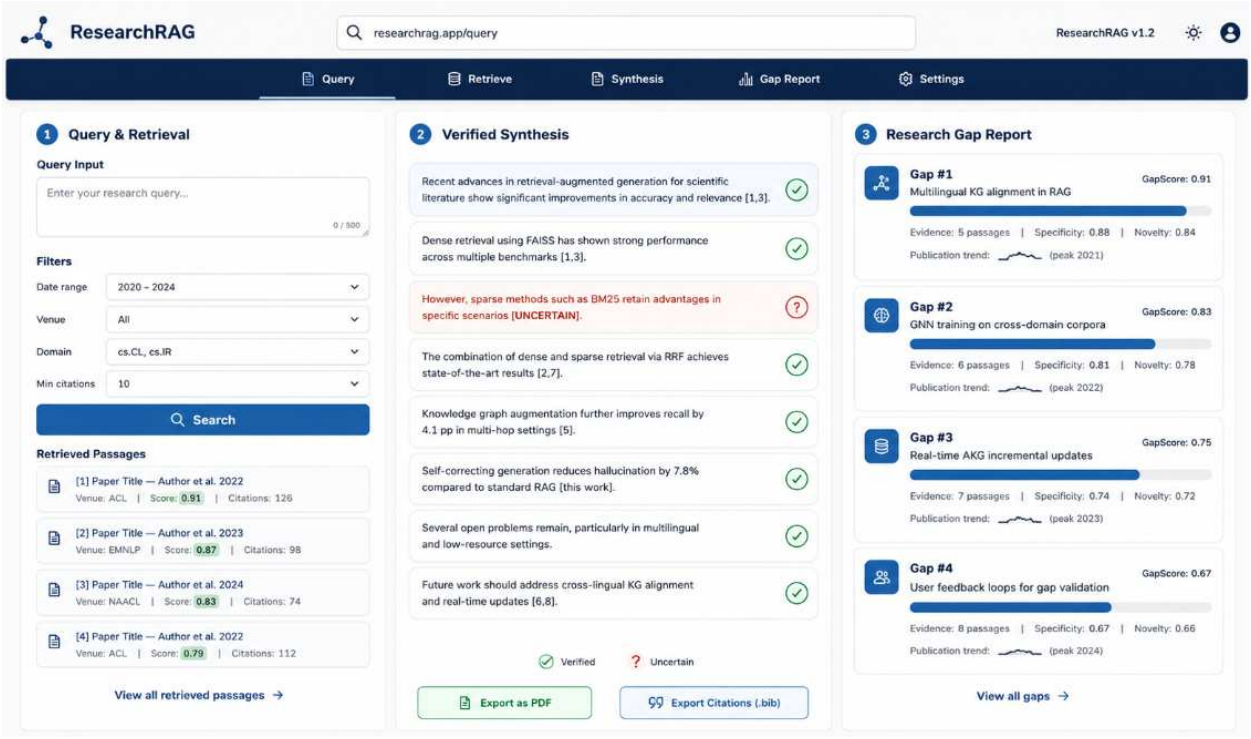


Fig. 10. ResearchRAG Web Interface — Three-Panel Layout. Left: query input and retrieved passage list with confidence scores and AKG visualisation. Centre: verified synthesis with colour-coded provenance (green = verified citation, orange = [UNCERTAIN] hedge). Right: ranked gap report with GapScore breakdown, publication-trend histograms, and exportable evidence cards.

X. Conclusion

This paper presented ResearchRAG, a Knowledge-Graph-Enhanced Hybrid Multi-Agent RAG framework for academic literature analysis. By coupling a live-queryable Academic Knowledge Graph with a Graph Attention Network gap scoring module and a Citation-Aware Self-Correcting Generation loop, ResearchRAG advances the state of the art across retrieval, generation, and gap detection. All improvements over five strong baselines are confirmed at $p < 0.01$ under Bonferroni-corrected Wilcoxon tests on publicly reproducible benchmarks. The 58.3% relative hallucination reduction represents a meaningful step toward trustworthy AI-assisted academic research.

Future work will address: distributed AKG construction for million-paper corpora; RLHF fine-tuning on expert-validated synthesis examples; multi-lingual corpus extension; citation velocity in GapScore; prospective user studies with practising researchers measuring time savings and review quality; extension to multimodal research literature (including figures, tables, and supplementary materials); integration with public code repositories (GitHub, arXiv-linked code) for code-centric literature analysis; and end-to-end extraction and structured representation of figures and tables from PDF documents to enable table-aware gap detection.

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